

Human Evaluation: Statistics and Process

Blind 3-way ranking of tutor policies — design memo

Study: Which teaching example helps most?
Tasks: VL (+ied past-tense inflection), MCL (order-of-operations misconceptions)
Sampling seed: 42
Status: Item & policy selection final; rating instrument live

1. Final numbers

	MCL (OOA)	VL (+ied)	Total
Items	20	30	50
Raters	4	4	8
Items / rater	10	15	—
Ratings / item	2	2	100 judgments

Session length scales with items: MCL sessions (10 items × a 3-way ranking) run ~15–20 minutes; VL sessions (15 items) run ~20–30 minutes — both reasonable for a single sitting on a crowdsourcing platform (Prolific / MTurk).

2. Tutor policy selection

Three example-selection policies are compared per task — a Best, a Middle, and a Worst anchor — scored by three metrics: BGI (behavioral-gain index), OTER (observed post-example error rate, from raw trajectory logs), and SR (success rate).

VL +ied — kept as-is

Level	Policy	BGI	OTER	SR
Best	llm_gemini31flashlite_nogold_freeform	0.0381	0.032	1.00
Middle	suffix_similarity	0.0285	0.065	1.00
Worst	llm_qwen8b_nogold_nonadaptive	0.0207	0.263	0.71

OTER is monotonic with BGI (0.032 → 0.065 → 0.263): both metrics agree on the ordering, so this trio is robust regardless of which effectiveness metric a reviewer weights most.

MCL OOA — Worst slot swapped to a traditional policy

Level	Policy	Type	BGI	OTER	SR
Best	llm_gpt5codex_nogold_nonadaptive	LLM	0.0272	0.507	0.362
Middle	llm_qwen8b_nogold_nonadaptive	LLM	0.0145	0.673	0.213
Worst	lemma_similarity	Traditional	0.0075	0.754	0.053

No traditional no-gold policy comes close to the LLM tutors' BGI — the best traditional policy tops out at 0.0075, below even the LLM Middle's 0.0145. Forcing one into Best or Middle would misrepresent it as competitive, so the Worst slot was swapped to traditional instead — consistent with §8.2's finding that traditional tutors struggle most under no-gold / MCL conditions. `lemma_similarity` was chosen over the near-identical `suffix_similarity` (BGI 0.0075 vs. 0.0073) only so the two task tables don't reuse the same policy name at different ranks — `suffix_similarity` is already VL's Middle. All three metrics are monotonic across this trio.

3. Item selection and validation

Task	Items	3 distinct policies	OOA cluster only
VL +ied	30	30/30	—
MCL OOA	20	20/20	20/20

Same random seed (42) and source files — sampling is exactly reproducible.

4. Rater assignment

Each task splits into two chunks; each chunk is rated independently twice, by a different rater per pass — the source of the "2 ratings / item" above.

Chunk	Task	Items	Raters (pass 1 / pass 2)	Item list
A	VL	15	1 / 5	angelify, bury, caddy, commify, esterify, heroify, hinny, mellify, misqualify, necropsy, coapply, dogsbody, epoxy, atrophy, babify
B	VL	15	2 / 6	phenocopy, pillory, plebify, premodify, queerify, recalcify, refry, rejustify, remarry, unnotify, misapply, travesty, colly, gentrify, misclassify
C	MCL	10	3 / 7	items 7, 45, 51, 200, 402, 470, 566, 580, 612, 676
D	MCL	10	4 / 8	items 722, 739, 840, 911, 1165, 1327, 1639, 1721, 1804, 1845

No rater sees both tasks; no two chunks share items. Pairing is 1↔5, 2↔6, 3↔7, 4↔8 — each pair rates its chunk's identical items independently.